

CAPE Chemistry

Unit 2

Titrimetric Analysis

Primary Standards and Standard Solutions

Primary Standards

The concentrations of all solutions used to analyze substances by titrimetric analysis must be accurately known. The concentrations of the hydrochloric acid and the sodium hydroxide solutions used in the analysis of vinegar, household cleaners, vitamin C tablets, aspirin and antacids would have been established by titrating these solutions against solutions of primary standards.

A compound used as a primary standard must meet the following criteria:

- The compound must be stable, so that it can be easily weighed under normal atmospheric conditions and not change during storage.
- The compound must be easy to obtain in highly purified form.
- It should dissolve readily in the titration medium in which it is to be used.
- It should have a high formula weight so that weighing errors are minimized.
- The reaction between the primary standard and the compound being standardized should be fast and quantitative (go to completion).
- The compound should not be expensive.

NB: Primary standard reagents are of about **99.99%** purity.

The commonly used primary standard reagents for the standardization of acids are:

- Sodium carbonate – Na_2CO_3
- Sodium hydrogen carbonate – NaHCO_3

The commonly used primary standard reagents for the standardization of permanganate solutions are:

- Oxalic acid – $(\text{COOH})_2$
- Sodium oxalate – $(\text{COONa})_2$

The commonly used primary standard reagent for the standardization of thiosulfate solutions is:

- Potassium iodate – KIO_3

Standard Solutions

A standard solution is a solution with an accurately known concentration.

When preparing a standard solution, the approach varies depending on whether the desired solution is made from a solid or a stock solution.

Preparation from a solid

Preparing 1.00 L of 0.100 M NaOH from solid

Calculation

- Moles required = $C \times V = 0.100 \text{ mol/L} \times 1.000 \text{ L} = 0.100 \text{ mol}$
- Molar mass NaOH = 40.00 g/mol
- Mass = $0.100 \text{ mol} \times 40.00 \text{ g/mol} = 4.00 \text{ g}$

Procedure

1. Dry/clean the weighing boat or use a tared weigh paper.
2. Accurately weigh **4.00 g** NaOH pellets (record mass). NOTE: NaOH is hygroscopic – weigh quickly.
3. Place about 200 mL of distilled water in a clean beaker. (Do not fill the 1 L flask yet)
4. Add the weighed NaOH pellets to the water **gradually** with stirring (or onto the stirrer). Dissolution is exothermic – solution will warm. Allow to cool if necessary.
5. Transfer the solution to a **1.00 L volumetric flask** using a funnel. Rinse the beaker and funnel with small portions of distilled water and add rinses to the volumetric flask to avoid loss.
6. Fill the flask to the calibration mark with distilled water at roughly room temperature (avoid large temperature differences). Add water dropwise as you approach the mark so the bottom of the meniscus sits on the mark.
7. Stopper and invert or gently mix the volumetric flask several times to ensure homogeneity.
8. Label the flask *0.100 M NaOH, 1.00 L, date, preparer, hazard (corrosive)*

Preparation from a Stock solution:

Preparing 1.00 L of 0.100 M HCl from 12.0 M stock

Calculation (dilution)

Use $C_1V_1 = C_2V_2$.

$$V_1 = (C_2V_2) / C_1$$

$$V_1 = (0.100 \text{ M} \times 1000 \text{ mL}) / 12.0 \text{ M}$$

$$V_1 = 8.333 \text{ mL}$$

So, measure 8.33 mL of 12.0 M HCl and dilute to 1.00 L

Procedure

1. Put on PPE and work in a fume hood if possible (conc. HCl fumes).
2. Rinse the pipette with a small amount of the stock HCl (optional practice step) and then draw **8.33 mL** of the 12.0 M HCl with the pipette. (If using a 10 mL pipette, record exact delivered volume and technique.)

3. Add about 300 mL of distilled water to a **1.00 L volumetric flask**. **Always add acid to water**, never water to acid.
4. Carefully dispense the measured 8.33 mL of conc. HCl into the volumetric flask containing the water. (Add the acid slowly, with the flask in the fume hood.)
5. Rinse the pipette into the flask with a small amount of distilled water to transfer any residual acid.
6. Allow solution to equilibrate then make up to the 1.00 L mark with distilled water, adding dropwise near the mark.
7. Stopper and invert/mix thoroughly.
8. Label: *0.100 M HCl, 1.00 L, prepared from 12.0 M stock (8.33 mL), date, preparer, hazard (corrosive)*